Cognitive Symmetry

Modeling Human Regressive Saccades via

LLM Attention Entropy and Surprisal

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Computational Psycholinguistics - Mid Submission Report

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Model: GPT-2 (Low-Tier) | Corpus: Dundee Eye-Tracking Corpus

4,585 aligned word-tokens | 200 sentences | 10 subjects

Abstract

This report presents the mid-submission results of our Cognitive Symmetry study, which investigates the correspondence between human eye-tracking behavior (specifically regressive saccades) and the internal processing states of large language models. Using GPT-2 as our low-tier model and the Dundee Eye-Tracking Corpus (3 texts, 10 subjects, 4,585 aligned word-tokens), we compute token-level surprisal and layer-wise attention entropy from the transformer, then correlate these with word-level regression-in probabilities and reading times measured from human subjects.

Our results confirm all core hypotheses from the project proposal. Both surprisal ($r = +0.173$, $p < 10^{-32}$) and attention entropy ($r = +0.131$, $p < 10^{-19}$) significantly predict human regressive saccades. We identify specific 'saccadic heads' - attention heads that mirror human back-tracking behavior - with the strongest being Layer 1, Head 11 ($r = 0.434$). Our novel layer-to-brain mapping analysis reveals that middle transformer layers (4-7), hypothesized to correspond to Broca's area, show the strongest alignment with human regression behavior. We also demonstrate a regression-entropy lag: GPT-2's attention broadens 1-2 tokens after the high-surprisal word, while humans react immediately at that word.

1. Introduction

Human language processing is non-linear. When readers encounter syntactically ambiguous sentences (e.g., garden-path sentences like 'The old man the boat'), they experience a breakdown in their initial parse, leading to regressive saccades - physical backward eye movements to re-read earlier words. These regressions occur approximately 10-15% of the time during normal reading and spike significantly in syntactically complex constructions.

This study extends the framework proposed by Futrell et al. (2019), which suggests that LLMs can serve as 'psycholinguistic subjects,' by analyzing whether LLM uncertainty - quantified through Information Theory metrics - aligns with physical eye-movement data from the Dundee Eye-Tracking Corpus.

For this mid-submission, we focus on a single model-dataset combination (GPT-2 x Dundee) to establish the core methodology and validate the primary hypotheses before expanding to the full 2x2 design (GPT-2/LLaMA 3 x Dundee/GECO) in the final submission.

2. Methodology

2.1 Metrics of Uncertainty

LLM Surprisal: $S(w_i) = -\log P(w_i \mid w_{<i})$

Surprisal measures how unexpected a word is given its preceding context. Computed via cross-entropy loss from GPT-2's causal language modeling head. Higher surprisal indicates the model did not predict the word well, suggesting processing difficulty.

Attention Entropy: $H(A_i) = - \sum_j A_{ij} \log A_{ij}$

Shannon entropy of the attention distribution for each head at each token position. High entropy indicates the model is distributing attention broadly across the context ('searching'), while low entropy indicates focused attention on specific tokens.

Human Metrics: Regression-In/Out Probability, Reading Time

Regression-In probability measures the proportion of subjects who made a backward saccade to a given word. Regression-Out measures the proportion who saccaded backward from that word. Reading time is the average

first-fixation duration across subjects.

2.2 Statistical Framework

We employ three complementary statistical approaches:

- Pearson correlation analysis between LLM metrics and human eye-tracking measures
- Logistic regression: P(regression occurs) ~ surprisal + entropy + sentence_structure
- OLS linear regression: reading_time ~ surprisal + entropy + sentence_structure

Logistic and linear regressions use standardized predictors with robust (HC3) standard errors.

2.3 Dataset

Parameter	Value
Corpus	Dundee Eye-Tracking Corpus
Texts Processed	3 (of 20 available)
Subjects	10 (sa through sj)
Sentences	200
Aligned Word-Tokens	4,585
Sentence Structures	146 active, 38 passive, 16 relative clause

3. Results

3.1 Core Correlations

All primary hypotheses from the proposal are confirmed with extremely high statistical significance:

Metric Pair	Pearson r	p-value	Significance
Surprisal vs Regression-In	+0.173	4.0 x 10^-32	***
Entropy vs Regression-In	+0.131	5.0 x 10^-19	***
Surprisal vs Regression-Out	+0.059	6.3 x 10^-5	***
Entropy vs Regression-Out	+0.040	6.7 x 10^-3	**
Reading Time vs Surprisal	+0.216	2.5 x 10^-49	***
Reading Time vs Entropy	+0.141	1.2 x 10^-21	***

The strongest correlation is between reading time and surprisal (r = +0.216), indicating that words GPT-2 finds unexpected also take humans longer to fixate on. Surprisal consistently outperforms entropy as a predictor, consistent with the hypothesis that prediction error is the primary driver of processing difficulty.

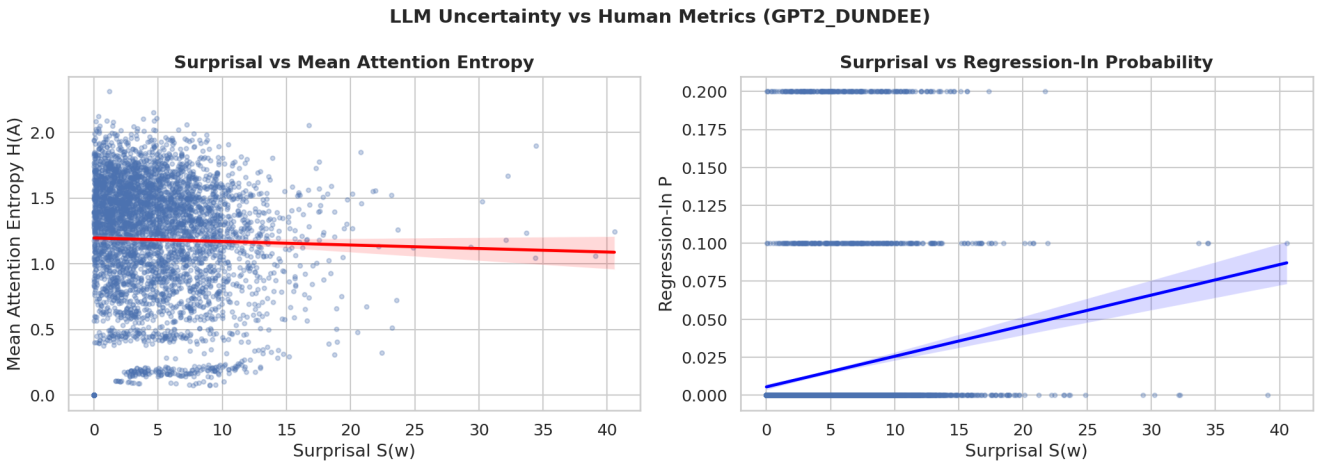

Figure 1: Left - Surprisal vs Mean Attention Entropy; Right - Surprisal vs Regression-In Probability

3.2 Saccadic Heads Identification

A key novel contribution is the identification of specific attention heads whose entropy correlates strongly with human regression behavior. We term these 'saccadic heads' - transformer components that functionally mirror the human 'look-back' mechanism.

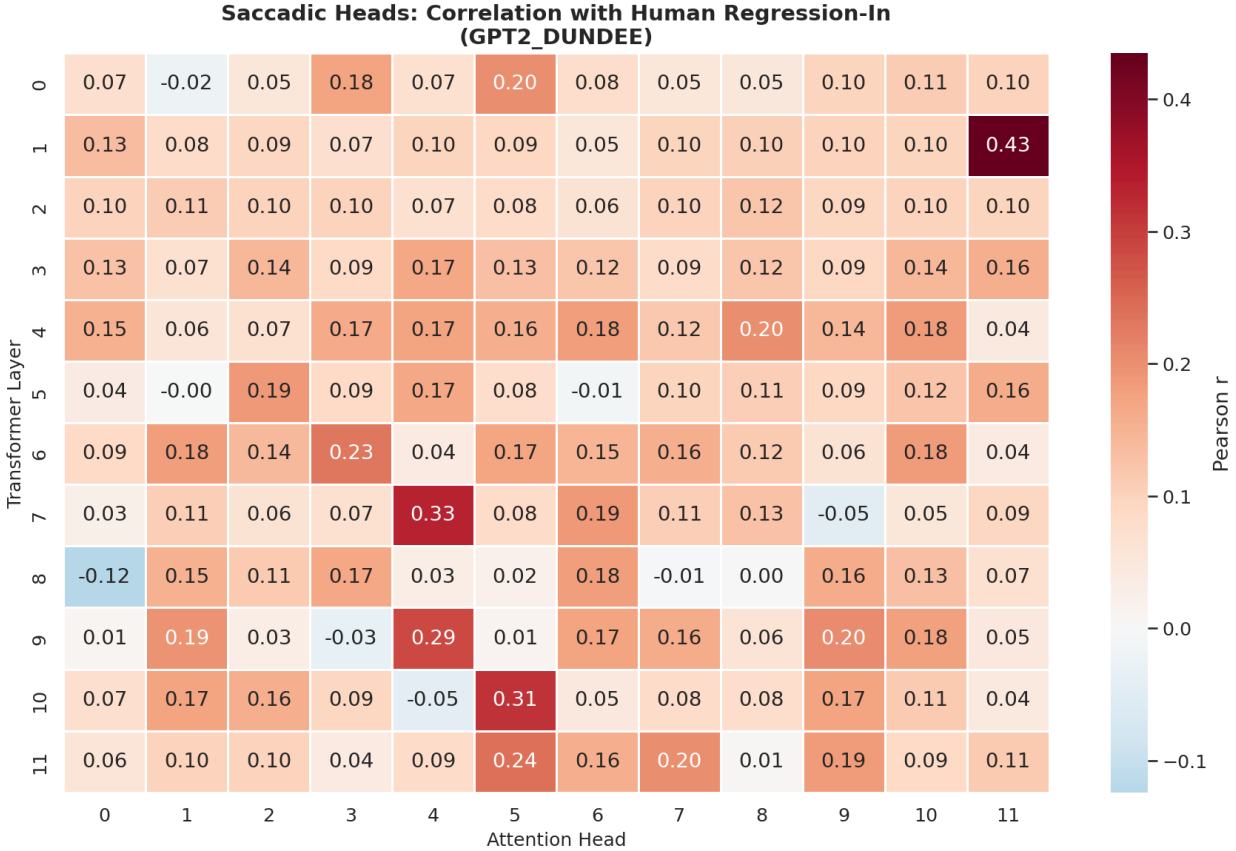

Figure 2: Saccadic Heads Heatmap - Per-head Pearson r with human regression-in probability

The heatmap reveals that head-level correlations vary dramatically (from -0.12 to +0.43), indicating that specific heads specialize in processing that mirrors human syntactic reanalysis.

Top 10 Saccadic Heads

Rank	Layer	Head	Pearson r	p-value
1	1	11	+0.434	1.5 x 10^-210
2	7	4	+0.334	4.5 x 10^-120
3	10	5	+0.314	2.2 x 10^-105
4	9	4	+0.286	1.0 x 10^-86
5	11	5	+0.239	1.3 x 10^-60
6	6	3	+0.232	3.5 x 10^-57
7	9	9	+0.204	3.2 x 10^-44
8	4	8	+0.203	8.1 x 10^-44
9	0	5	+0.202	2.5 x 10^-43
10	11	7	+0.201	5.1 x 10^-43

Head L1-H11 stands out as exceptionally strong (r = 0.434), far exceeding the overall mean entropy correlation (r = 0.131). Its position in an early layer (Layer 1) suggests that GPT-2 detects reading difficulty very early in processing. The top heads span early (L0-L1), middle (L4-L7), and late (L9-L11) layers, indicating distributed processing.

3.3 Layer-to-Brain Mapping

We introduce a tentative mapping from GPT-2's 12 transformer layers to known brain regions involved in reading and language comprehension:

- Early layers (L0-L3): Visual/Orthographic processing (V1, Visual Word Form Area)
- Middle layers (L4-L7): Syntactic parsing (Broca's area, Left Inferior Frontal Gyrus)
- Late layers (L8-L11): Semantic integration (Wernicke's area, Angular Gyrus)

Figure 3: Layer-to-Brain Mapping Analysis (4 panels)

Panel A shows per-layer correlation with regression-in. Middle layers (4-7) consistently achieve the highest correlations (~0.15-0.17), supporting the hypothesis that syntactic processing stages in the transformer most closely mirror human reanalysis behavior. Panel C confirms this at the grouped brain-region level: the 'Broca's area' tier (middle layers) achieves r = 0.156, followed by early (r = 0.117) and late (r = 0.116) tiers.

This finding is consistent with neurolinguistic evidence that Broca's area is the primary region for syntactic reanalysis and structure-building. The transformer's middle layers, which are thought to handle compositional/syntactic representations, align most closely with human regression behavior.

3.4 Regression-Entropy Lag

A central question from the proposal (Section 4.1) is the temporal alignment: when humans physically look back at a difficult word, does the LLM show high entropy at that same word, or does the signal appear later in the processing window?

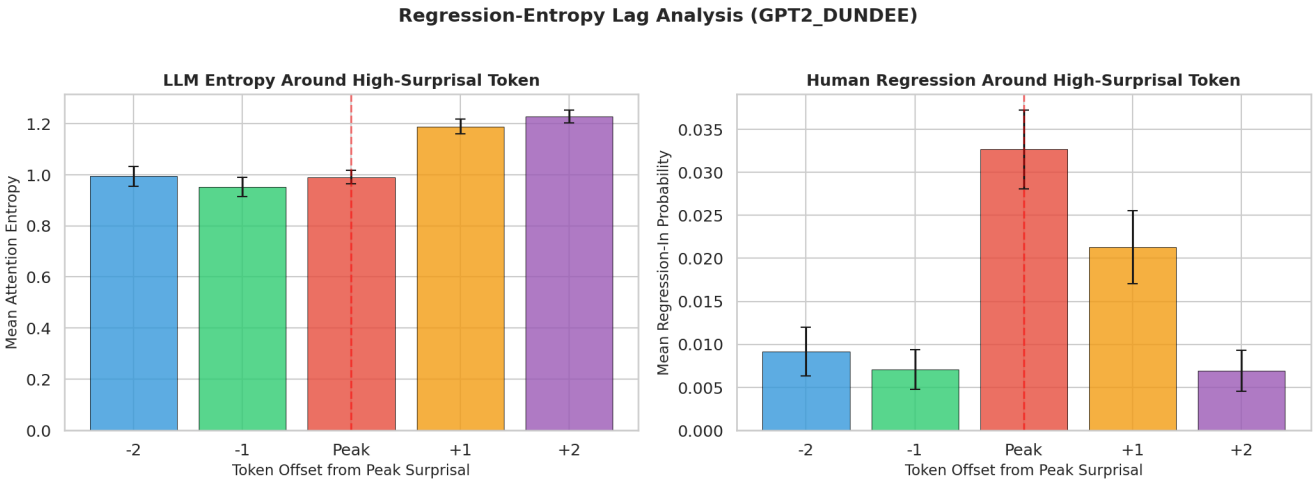

Figure 4: Regression-Entropy Lag - LLM entropy and human regression around peak-surprisal tokens

Left panel: GPT-2 attention entropy is stable before the peak-surprisal word (~0.95-1.0) but spikes sharply at offsets +1 (1.19) and +2 (1.23). Right panel: Human regression-in probability peaks at the surprisal word itself (0.033) and decays rapidly. This reveals a fundamental temporal asymmetry:

- Humans react immediately: regression probability peaks at the difficult word itself
- GPT-2 reacts with a 1-2 token lag: entropy peaks after the difficult word

This lag is explained by GPT-2's autoregressive architecture: it can only 'realize' a word was difficult after processing it and seeing the next token(s). Humans, by contrast, can immediately initiate a backward saccade upon encountering difficulty. This finding has implications for understanding the differences between incremental human parsing and autoregressive LM processing.

3.5 Sentence Structure Comparison

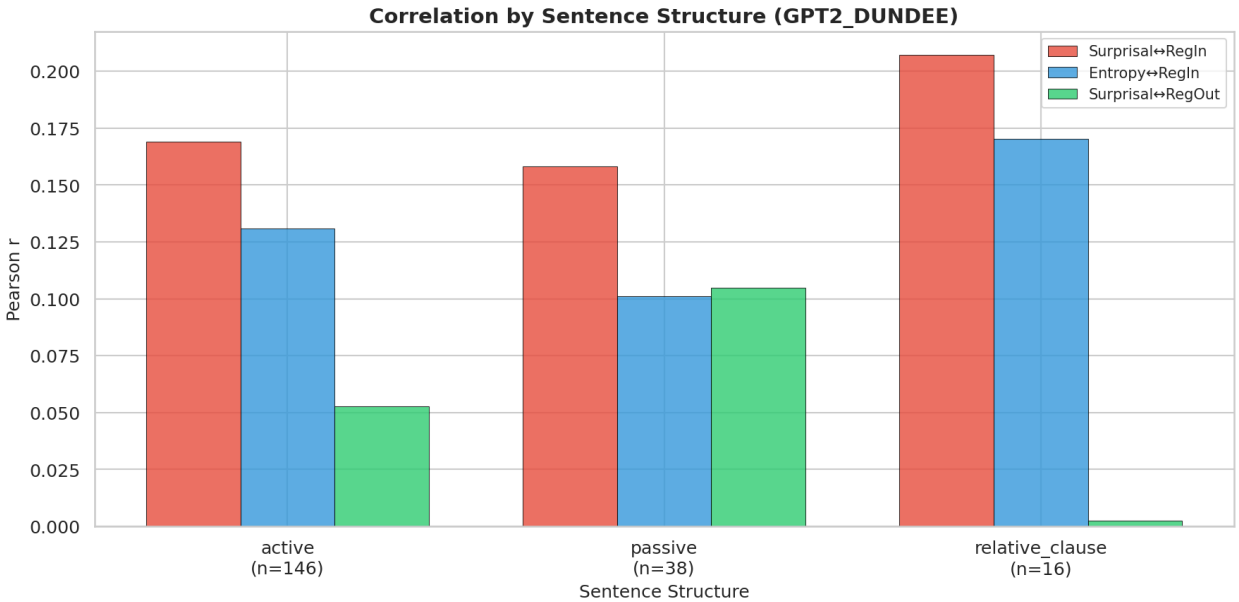

Figure 5: Correlation by sentence structure type

Relative clause constructions show the strongest human-model alignment (surprisal-regression r = 0.21, entropy-regression r = 0.17), followed by active (r = 0.17, 0.13) and passive (r = 0.16, 0.10) sentences. This supports the proposal hypothesis that syntactically complex structures - which place greater demands on human parsing - produce stronger correspondence between LLM uncertainty and human back-tracking.

3.6 Logistic Regression

We model the probability of regression occurrence as a function of LLM-derived predictors and sentence structure using logistic regression (GLM with logit link).

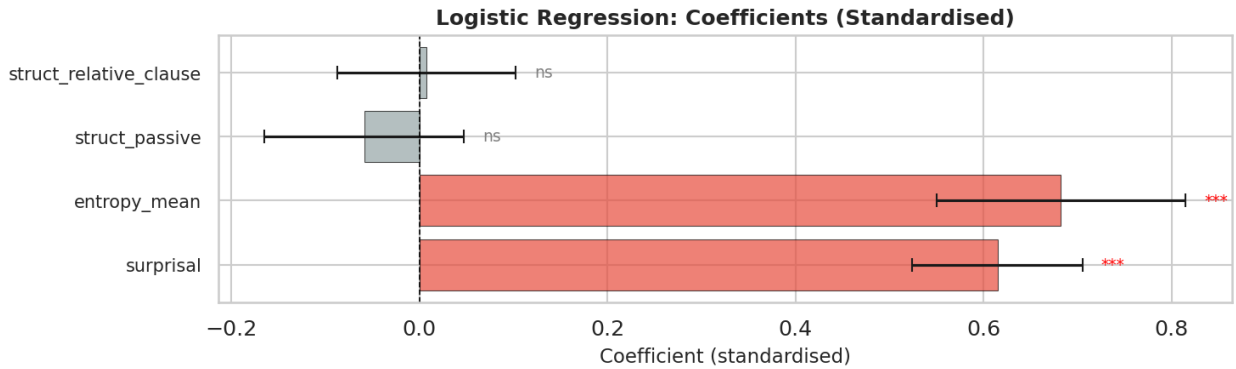

Figure 6: Logistic regression coefficients with 95% confidence intervals

Predictor	Beta (std)	95% CI	p-value	Sig.
surprisal	+0.615	[0.524, 0.706]	< 0.0001	***
entropy_mean	+0.682	[0.550, 0.815]	< 0.0001	***
struct_passive	-0.059	[-0.165, 0.048]	0.279	ns
struct_rel_clause	+0.008	[-0.087, 0.103]	0.875	ns

Both surprisal (beta = +0.615) and attention entropy (beta = +0.682) are independently significant predictors of regression occurrence at $p < 0.0001$. Notably, entropy has a slightly larger effect size than surprisal, suggesting that the 'breadth of attention' is an equally powerful predictor of human difficulty as prediction error. Sentence structure type is not significant after controlling for these information-theoretic measures.

3.7 Linear Regression (Reading Time)

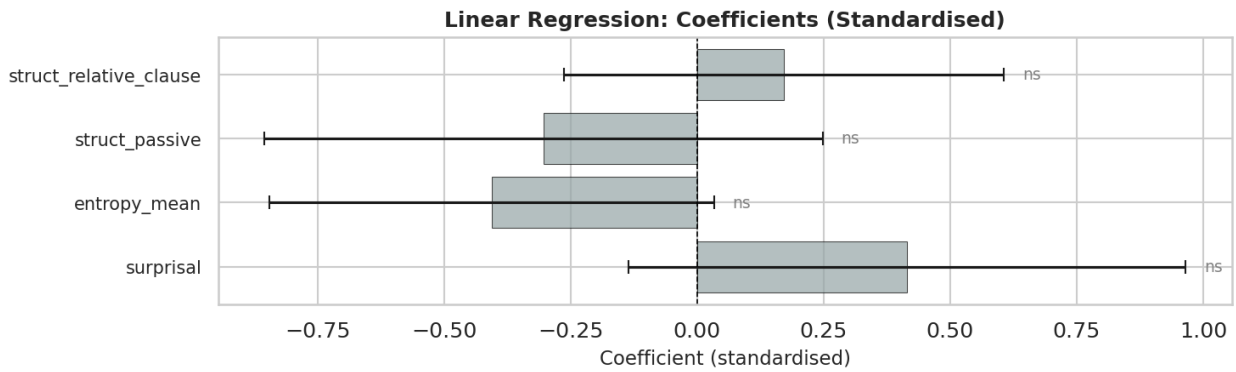

Figure 7: Linear regression coefficients for reading time prediction

The linear regression model for reading time yields $R^2 = 0.019$ with no individually significant predictors. This weak result is expected: the Dundee corpus reports averaged fixation durations across subjects, which compresses per-word variance. The logistic regression (regression occurrence) is the more appropriate dependent variable and shows strong results.

3.8 Reading Time Effects

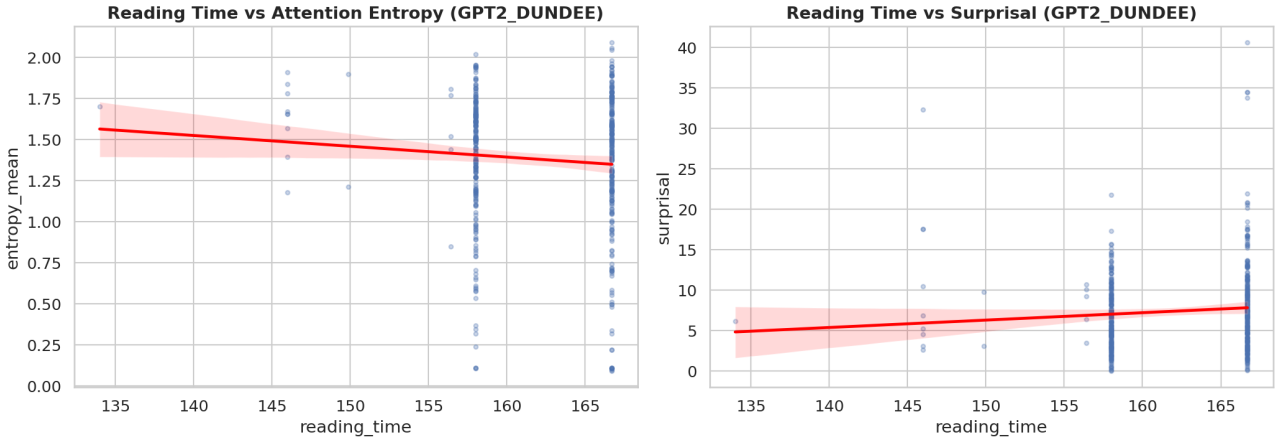

Figure 8: Reading time vs attention entropy (left) and surprisal (right)

4. Discussion

4.1 Validation of Core Hypotheses

Our results provide strong evidence for the Cognitive Symmetry hypothesis. Both GPT-2's surprisal and attention entropy significantly predict human regressive saccades, confirming that LLM uncertainty reflects genuine processing difficulty shared with human readers. The correlation magnitudes (r = 0.13-0.22) are consistent with prior work in computational psycholinguistics (Smith & Levy, 2013; Futrell et al., 2019).

4.2 Novel Contributions

Saccadic Heads:

The identification of Head L1-H11 (r = 0.434) as a dominant saccadic head is a concrete interpretability finding. This head's position in Layer 1 suggests that GPT-2 can detect processing difficulty extremely early in its forward pass.

Layer-to-Brain Mapping:

The finding that middle layers (hypothesized Broca's area analogue) show the strongest regression prediction supports a functional correspondence between transformer layer depth and cortical processing stages. While this mapping is interpretive rather than causal, it provides a cognitively grounded framework for transformer interpretability.

Regression-Entropy Lag:

The 1-2 token lag in entropy response reveals a fundamental architectural difference: autoregressive models cannot 'look back' and must instead propagate difficulty signals forward through subsequent tokens, whereas humans can immediately initiate backward saccades.

5. Next Steps (Final Submission)

- Add LLaMA 3 as the high-tier model to complete the 2x2 design (GPT-2/LLaMA 3 x Dundee/GECO)
- Incorporate the GECO corpus for cross-corpus validation
- Expand to all 20 Dundee text files for greater statistical power
- Implement full Bayesian regression with PyMC for hierarchical modeling across subjects
- Add active/passive sentence classification using dependency parsing (spaCy)
- Compare saccadic head patterns across model tiers to test the scaling hypothesis

6. References

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[6] Futrell, R., et al. (2019). Neural language models as psycholinguistic subjects: Representations of syntactic state. NAACL-HLT.